



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

RA6M5-Based IoT Development Board with BLE, USB-C, Battery Management, QSPI Flash, microSD, and Qwiic Interface

**The domain of the Project:
G5-PCB Design**

Team Mentors:

**Mr. Paramesh
(Mentor SURE Trust)**

Team Members:

Mr. KOLIMI MOHAMMED HARSHAD VALI

Period of the project

March 2026 to April 2026



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Declaration

The project titled “**RA6M5-Based IoT Development Board with BLE, USB-C, Battery Management, QSPI Flash, microSD, and Qwiic Interface**” has been mentored by -----, organised by SURE Trust, from April 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:-

Mr. KOLIMI MOHAMMED HARSHAD VALI

Signature:-

Mentor's Name:- Mr.Paramesh

Designation:- Mentor -SURE Trust

Executive Director:-

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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Executive Summary

- A concise overview of the project's objectives, methods, key findings, and recommendations.
- Highlights the most important aspects of the project for executives or stakeholders.

The objective of this project was to design a professional embedded hardware development board using the Renesas RA6M5 microcontroller. The board was developed to support modern IoT (Internet of Things) applications by integrating multiple hardware components into a single PCB.

The hardware includes USB Type-C power input, battery charging and protection, Bluetooth Low Energy (BLE) communication, QSPI Flash memory, microSD card support, Qwiic connector for sensors, SWD programming interface, status LEDs, and voltage regulation circuits.

The complete schematic and a four-layer PCB were designed using KiCad software. The design follows standard PCB design practices, including proper component placement, signal routing, power distribution, and Design Rule Check (DRC). The project helped improve practical knowledge in PCB design, embedded systems, hardware architecture, and electronic circuit design.



Introduction

- Background and context of the project
- Problem statement or goals of the project
- Scope and limitations of the project
- Innovation component in the project

Background:-

Embedded systems are used in almost every modern electronic device such as smart home systems, industrial automation, medical equipment, automotive electronics, and IoT devices. These systems require reliable hardware that can process data, communicate with external devices, and consume very little power.

To understand how such systems are developed in industries, this project focuses on designing a complete embedded development board from scratch.

Problem statement:-

Many IoT applications require several external modules such as Bluetooth communication, memory storage, battery charging, and sensor interfaces. Designing all these modules separately increases system complexity and cost.

The aim of this project is to integrate all these required modules into one compact and reliable PCB.

Scope of the Project:-

The project includes:

- Designing the complete schematic.
- Selecting suitable electronic components.
- Creating a four-layer PCB.
- Designing power supply circuits.
- Integrating communication modules.
- Providing external memory support.
- Preparing manufacturing files such as Gerber and BOM.

The project does not include firmware programming or hardware testing after PCB fabrication.



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Innovation in the Project:-

Unlike simple development boards, this design integrates several industrial-level features into a single board.

These include:

- USB Type-C power input
- Battery charging circuit
- Battery fuel gauge
- Bluetooth Low Energy
- External QSPI Flash
- microSD storage
- Qwiic connector
- SWD programming interface

This makes the board suitable for real-world embedded and IoT applications.



Project Objectives

- Clearly defined objectives and goals of the project
- Expected outcomes and deliverables.

Project Objectives:-

The main objectives of this project are:

1. To design a complete embedded development board based on the Renesas RA6M5 microcontroller.
2. To create the complete schematic using KiCad.
3. To design a 4-layer PCB following industry-standard PCB design practices.
4. To integrate essential peripherals such as USB Type-C, Bluetooth Low Energy (BLE), QSPI Flash, microSD card, and Qwiic connector into a single board.
5. To design a reliable power management system, including battery charging and voltage regulation.
6. To ensure the PCB design is free from errors by performing Electrical Rule Check (ERC) and Design Rule Check (DRC).
7. To generate manufacturing files such as Gerber files and Bill of Materials (BOM) for PCB fabrication.
8. To improve practical knowledge and hands-on experience in embedded hardware design and PCB development.

Expected Outcomes:-

- A complete and error-free schematic.
- A professional 4-layer PCB layout.
- Manufacturing-ready Gerber and BOM files.
- A compact development board suitable for embedded and IoT applications.
- Enhanced skills in PCB design, component selection, and hardware development.



Methodology and Results

- Methods/Technology used
- Tools/Software used
- Data collection approach (if applicable)
- Project Architecture (Clearly describe the architecture)
- Final project working screenshots along with supporting explanation
- Project GitHub Link

Methodology:-

The project was carried out using a systematic approach to design a professional embedded development board. Each stage was completed carefully to ensure that the final PCB design was reliable, manufacturable, and suitable for IoT and embedded system applications.

Step 1: Requirement Analysis

The project requirements were identified, including USB Type-C power, battery charging, Bluetooth communication, external memory, and a stable power supply.

Step 2: Component Selection

Suitable electronic components were selected based on compatibility, performance, and application requirements. The main components include the Renesas RA6M5 microcontroller, BLE module, QSPI Flash, microSD card, battery charger, and voltage regulators.

Step 3: Schematic Design

The complete circuit was designed in **KiCad** by dividing it into functional blocks such as the power supply, microcontroller, communication, memory, and programming circuits. An **Electrical Rule Check (ERC)** was performed to verify the schematic.

Step 4: PCB Design

A **4-layer PCB** was designed in **KiCad** with proper component placement, signal routing, power planes, and ground planes to improve performance and reliability.

Layer Configuration:-

- Layer 1 – Components and Signal Routing
- Layer 2 – Ground Plane
- Layer 3 – Power Plane
- Layer 4 – Bottom Signal Routing



Step 5: Design Verification

The PCB was verified using **Design Rule Check (DRC)** to ensure there were no routing or manufacturing errors. All identified issues were corrected.

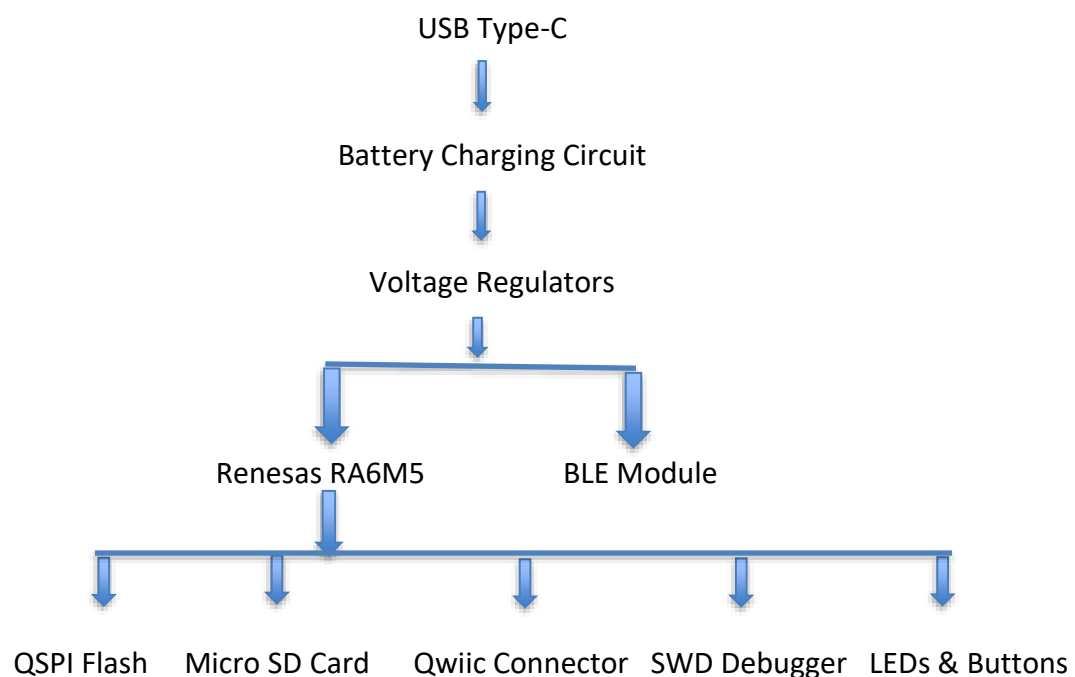
Step 6: Manufacturing Output

After successful verification, the required manufacturing files, including **Gerber files** and the **Bill of Materials (BOM)**, were generated for PCB fabrication.

Tools and Software Used:- KICAD

The project was designed using **KiCad**, an open-source Electronic Design Automation (EDA) software. KiCad was used for schematic design, PCB layout, design verification (ERC and DRC), 3D visualization, and generating Gerber and BOM files required for PCB manufacturing.

Project Architecture:-

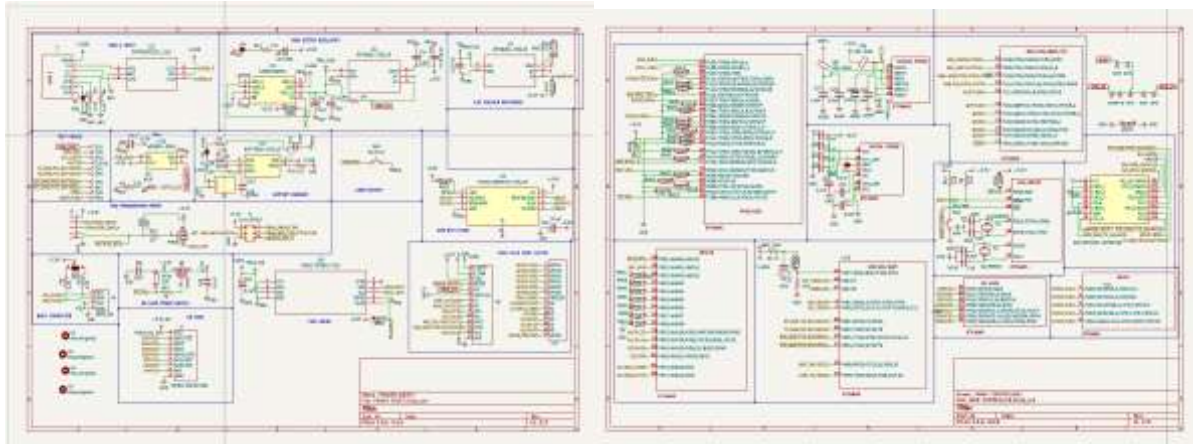




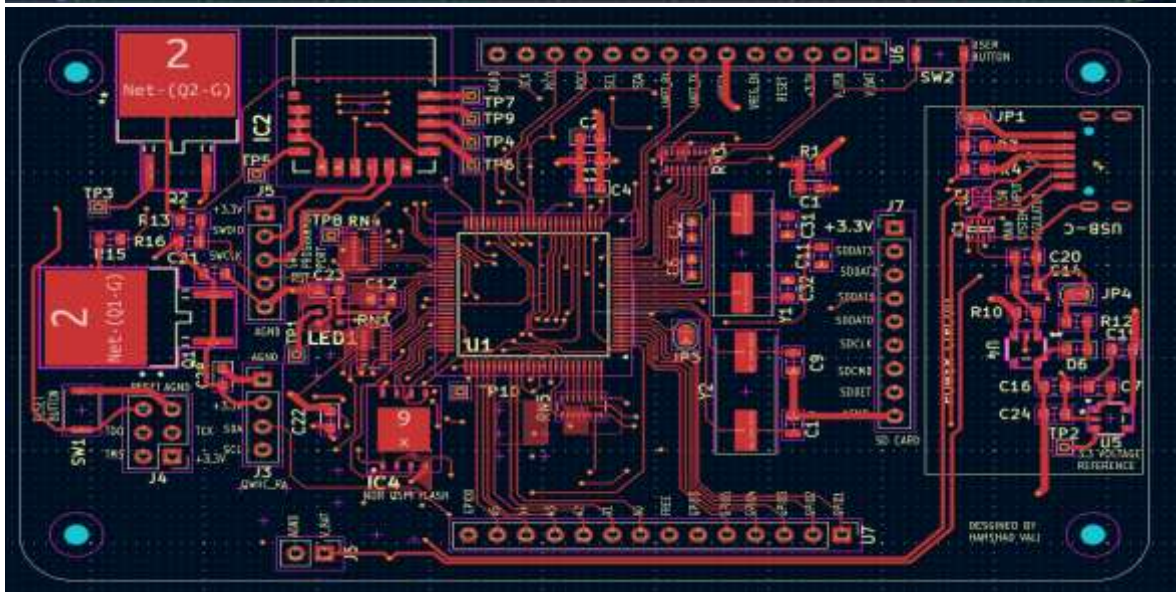
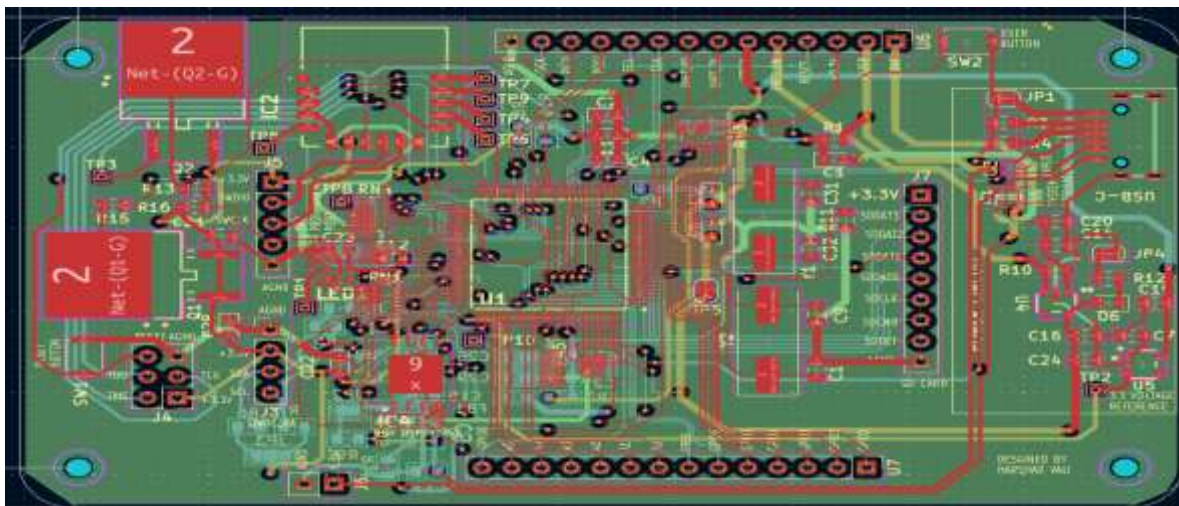
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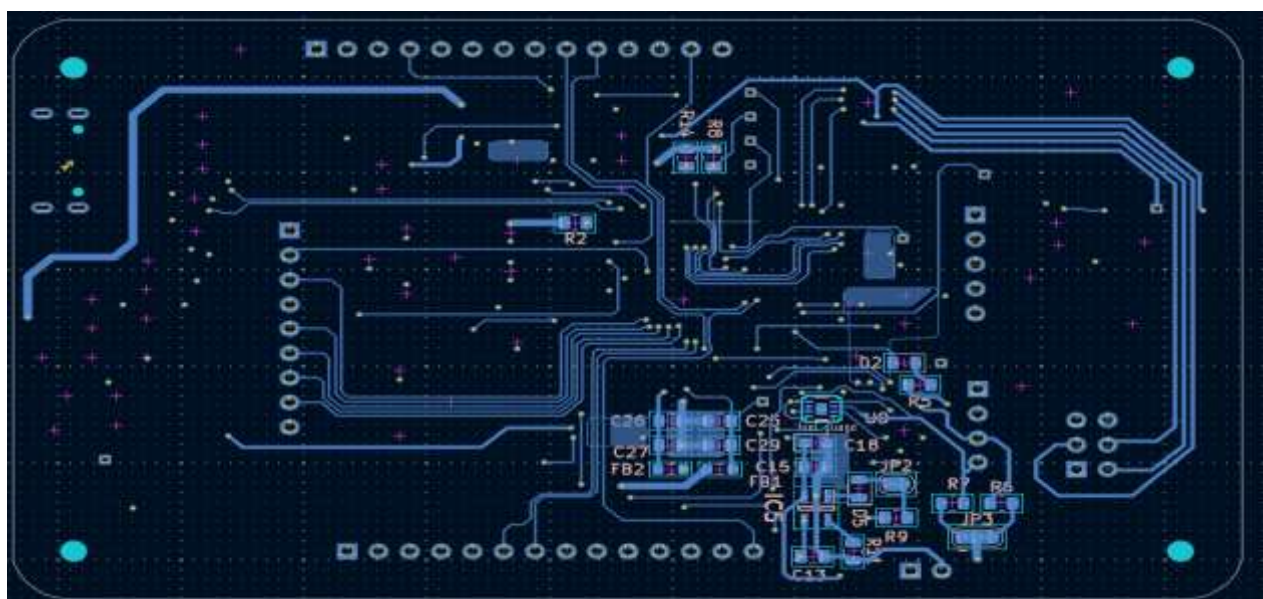
Final Project Screenshots:-

Complete Schematic Diagram:-



PCB Layout:-

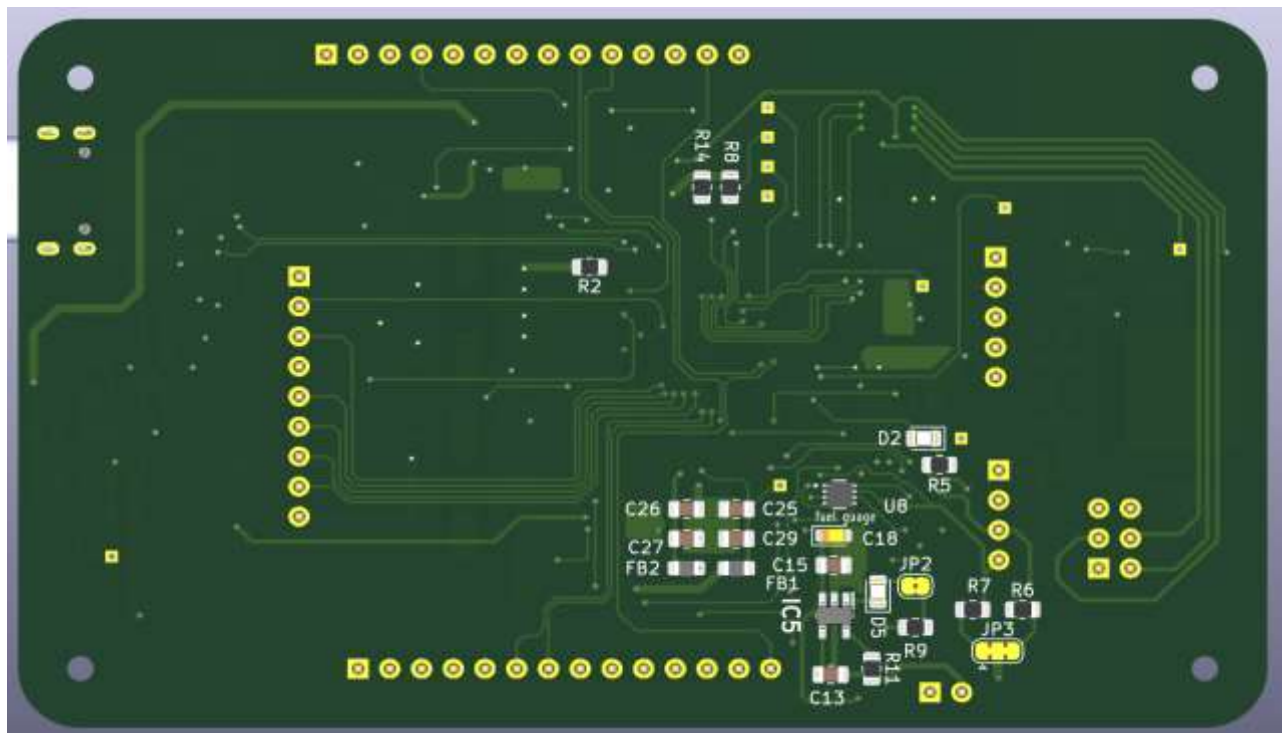
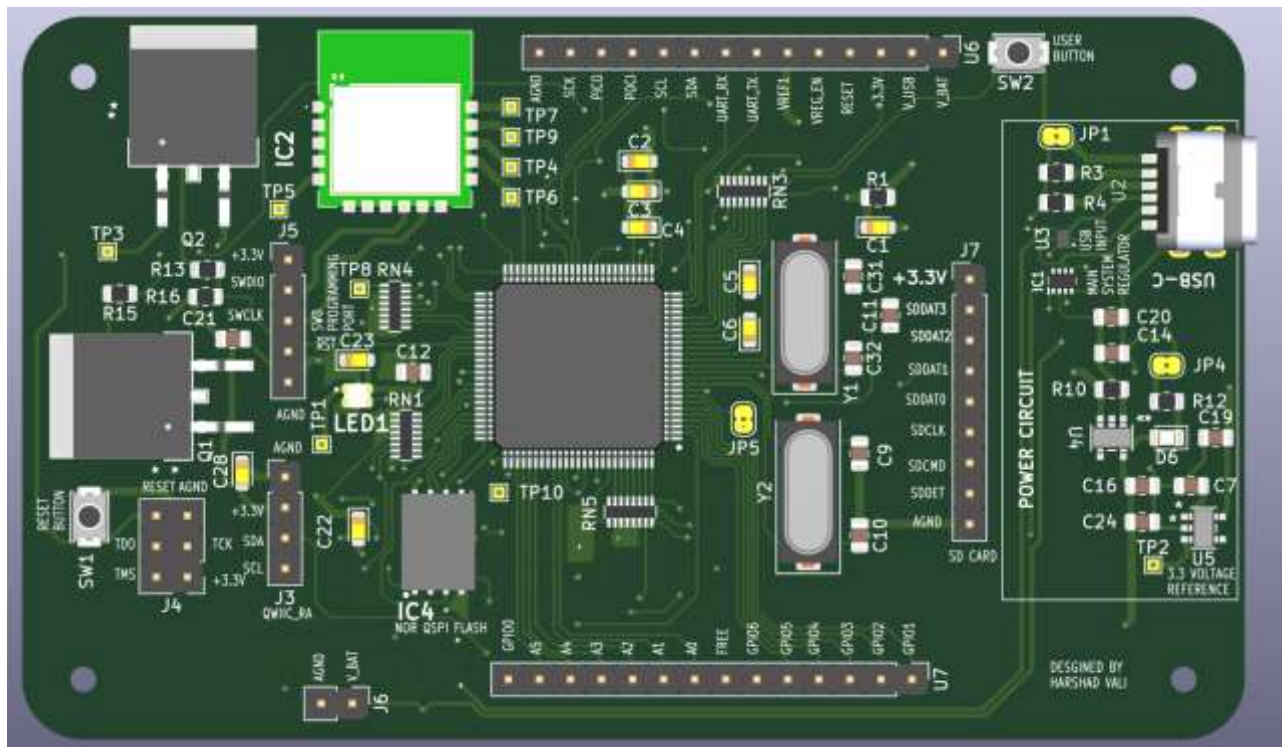






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3D View of the PCB:-





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Learning and Reflection

- Document every team member's new learnings (in terms of technology, management, etc)
- Document every team member's experience. (overall experience)

Learning:-

During this project, I gained practical knowledge in embedded hardware design and PCB development. I learned how to design electronic circuits using KiCad, create a 4-layer PCB, and follow industry-standard PCB design practices.

Some of the key learnings include:

- Understanding electronic components and their applications.
- Designing complete schematics using KiCad.
- Creating a professional 4-layer PCB layout.
- Proper component placement and signal routing.
- Designing power supply and voltage regulation circuits.
- Integrating USB Type-C, BLE, QSPI Flash, microSD card, and other peripherals.
- Performing Electrical Rule Check (ERC) and Design Rule Check (DRC).
- Generating Gerber files and Bill of Materials (BOM) for PCB manufacturing.

Reflection:-

Working on this project gave me hands-on experience in designing an embedded development board from start to finish. Initially, I faced challenges in component selection, PCB routing, and resolving ERC/DRC errors. By referring to datasheets, learning from mentors, and practicing regularly, I was able to overcome these challenges.

This project improved my technical knowledge, problem-solving ability, and confidence in PCB design. It also helped me understand the complete hardware design process used in the electronics industry. Overall, the project was a valuable learning experience and strengthened my interest in embedded systems and hardware development.

Skills Gained:-

- PCB Design using KiCad
- Schematic Design
- 4-Layer PCB Layout
- Electronic Component Selection
- Power Supply Design
- Design Verification (ERC & DRC)
- Gerber & BOM Generation
- Problem-Solving
- Technical Documentation



Conclusion and Future Scope

- Recap objectives and achievements
- Future scope of this project

Conclusion:-

The main objective of this project was to design a professional embedded development board using the **Renesas RA6M5 microcontroller**. The project was successfully completed by designing the complete schematic and a **4-layer PCB** using **KiCad**. Various hardware modules, including **USB Type-C, Bluetooth Low Energy (BLE), QSPI Flash memory, microSD card, battery charging circuit, voltage regulators, and SWD programming interface**, were successfully integrated into the design.

The PCB design was verified using **Electrical Rule Check (ERC)** and **Design Rule Check (DRC)**, ensuring that the design is reliable and ready for manufacturing. This project enhanced my practical knowledge of embedded systems, PCB design, and electronic hardware development while providing valuable hands-on experience with industry-standard design practices.

Future Scope:-

The project can be further improved by implementing the following features:

- Develop firmware for the Renesas RA6M5 microcontroller.
- Add support for **Wi-Fi** and other wireless communication technologies.
- Interface additional sensors using the Qwiic connector.
- Connect the board to cloud platforms for IoT applications.
- Optimize power consumption for battery-operated devices.
- Manufacture and test the PCB to validate hardware performance.
- Design a protective enclosure for practical deployment.
- Enhance the board with additional peripherals based on application requirements.